

GLENWOOD, MN, USA

WMO: 726547

Lat: 45.644N

Lon: 95.320W

Elev: 425

StdP: 96.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.4	-24.1	-31.1	0.2	-27.0	-27.8	0.3	-23.9	14.0	-11.7	12.3	-10.2	3.7	290	0.637

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	32.0	22.7	29.9	21.9	28.0	20.6	24.7	29.3	23.6	27.9	22.5	26.5	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.7	27.3	22.3	17.9	26.7	21.2	16.7	25.4	77.3	29.3	72.7	28.1	68.4	26.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	11.0	9.5	-30.8	34.6	3.6	1.6	-33.3	35.8	-35.4	36.7	-37.4	37.6	-40.0	38.8		
(5)			-30.9	26.6	3.4	1.2	-33.4	27.4	-35.4	28.1	-37.3	28.8	-39.8	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-11.5	-9.0	-1.9	6.5	13.6	19.2	21.8	20.4	16.1	8.2	-0.3	-8.0		
	DBStd	12.94	7.69	7.33	7.13	5.75	4.87	3.58	3.18	3.04	4.61	5.22	6.26	6.81		
	HDD10.0	2700	667	532	373	132	21	0	0	0	5	98	313	559		
	HDD18.3	4679	926	766	628	356	160	32	8	14	96	317	561	817		
	CDD10.0	1361	0	0	3	27	133	276	367	322	188	42	3	0		
	CDD18.3	296	0	0	0	1	13	57	116	78	29	2	0	0		
	CDH23.3	2613	0	0	0	15	170	495	1018	629	257	28	1	0		
	CDH26.7	766	0	0	0	4	51	133	332	171	68	6	0	0		
(14) Wind	WSAvg	4.5	4.7	4.7	4.8	5.2	5.1	4.2	3.6	3.6	4.2	4.7	4.8	4.6		
(15) Precipitation	PrecAvg	669	15	15	34	59	73	107	98	83	78	64	28	15		
	PrecMax	922	47	33	87	162	138	190	202	187	145	167	102	41		
	PrecMin	471	1	3	11	18	8	49	21	11	26	11	0	3		
	PrecStd	107	11	8	20	35	30	38	44	40	38	42	25	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	9.9	19.0	25.1	31.9	32.5	34.1	32.7	32.0	27.3	18.7	7.8		
		MCWB	2.7	7.9	13.1	14.1	18.8	22.9	24.2	23.1	21.5	16.7	12.6	4.6		
	2%	DB	2.4	5.9	14.1	22.1	27.6	30.0	32.2	30.2	28.6	22.4	14.0	4.1		
		MCWB	1.1	3.6	10.1	12.5	17.2	21.5	23.7	22.3	20.8	15.1	8.5	2.1		
	5%	DB	1.0	2.7	11.0	18.7	24.7	27.8	30.0	28.0	26.5	18.8	11.1	2.4		
		MCWB	0.2	1.5	8.0	10.8	15.5	19.8	22.9	21.0	19.1	13.4	7.7	1.2		
	10%	DB	-1.2	1.3	7.4	16.2	22.3	26.3	28.0	27.2	23.9	16.3	7.8	1.1		
		MCWB	-1.4	0.2	5.1	9.3	14.7	18.9	21.4	20.5	18.2	11.8	5.2	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	7.3	14.1	16.2	21.6	25.0	26.8	25.7	23.9	19.4	14.0	5.5		
		MCDB	4.8	9.4	16.8	21.9	27.2	30.1	31.7	30.5	28.5	23.4	17.5	7.8		
	2%	WB	1.2	3.5	10.9	13.8	19.3	23.3	25.3	23.9	22.3	16.5	10.2	2.7		
		MCDB	2.2	5.1	13.8	19.6	24.2	27.8	29.8	28.1	26.4	20.3	13.7	4.2		
	5%	WB	-0.1	1.9	7.7	11.8	17.5	21.7	24.1	22.7	20.9	14.1	7.6	1.3		
		MCDB	0.7	2.8	10.2	16.9	21.8	25.8	28.4	26.6	24.6	18.3	10.2	2.2		
	10%	WB	-1.7	0.2	5.0	10.0	15.7	20.2	23.0	21.5	19.4	12.3	5.4	0.0		
		MCDB	-1.4	1.2	7.6	15.1	20.6	24.6	26.9	25.2	22.9	15.6	8.1	0.8		
(35) Mean Daily Temperature Range	MDBR		9.0	9.4	9.2	11.1	11.4	10.8	11.0	11.1	11.3	10.0	8.5	8.0		
	5% DB	MCDBR	9.6	9.8	13.5	16.4	15.6	13.2	12.9	12.8	13.9	14.6	13.0	9.8		
		MCWBR	8.2	8.1	8.9	8.7	7.5	6.4	6.2	6.3	7.0	8.0	8.5	8.0		
	5% WB	MCDBR	9.1	9.3	12.7	14.5	12.9	11.1	11.6	11.4	11.7	12.6	12.0	8.9		
		MCWBR	7.9	7.8	8.8	8.9	7.7	6.8	6.8	6.9	7.1	8.1	8.5	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.280	0.307	0.357	0.383	0.384	0.382	0.390	0.356	0.309	0.277	0.253		
	taud		2.415	2.359	2.404	2.333	2.318	2.369	2.374	2.343	2.423	2.539	2.531	2.456		
	Ebn at Noon		863	906	928	902	884	883	880	858	862	866	841	837		
	Edh at Noon		70	90	100	118	124	119	117	116	98	76	63	60		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.80	3.80	4.73	5.40	5.99	6.41	5.34	4.07	2.56	1.63	1.19		
	RadStd		0.22	0.32	0.35	0.40	0.41	0.39	0.23	0.44	0.30	0.37	0.16	0.15		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page